PHASE 03 / PROJECT DOCUMENTATION

Project Design

Structuring the analytical solution before implementation.

The design translates the project's questions into a layered workflow and a dashboard experience that is consistent, interpretable, and grounded in the selected dataset.

01 | Design principle

The dashboard should prioritize decisions over decoration. Every page or chart must answer a specific question, use a suitable visual form, and make its scope clear. The design is flexible: a chart is included only if the selected dataset contains the necessary fields with sufficient quality.

02 | Proposed architecture

Layer	Role	Deliverable
Data input	Load the chosen CSV/Excel or approved source	Source note and data dictionary
Preparation	Inspect data types, missingness, duplicates, labels, and units	Clean analysis-ready table
Metrics	Define totals, averages, rankings, and valid shares	Metric definitions and checked calculations
Visualization	Build charts, filters, tooltips, and dashboard/story pages	Interactive analytical interface
Documentation	Record assumptions, screenshots, findings, and limits	Repository documentation

03 | Dashboard information design

MARKET OVERVIEW

Headline measures, data coverage, and a compact summary of the selected period.

MANUFACTURER & SEGMENT

Compare recorded sales or other available performance measures by company and vehicle category.

BATTERY & RANGE

Explore battery chemistry and compare capacity or range with clear units.

GEOGRAPHY & RATINGS

Inspect manufacturing-country distribution and customer ratings where reliable fields exist.

04 | Visualization mapping

Analytical task	Preferred visual	Design consideration
Compare manufacturers	Sorted horizontal bars	Limit clutter; show units and scope
Observe change over years	Line chart	Use a comparable yearly measure and avoid mixing cumulative with annual data
Compare categories and battery types	Grouped/stacked bars	Keep category colors consistent and labels readable
Study range/capacity variation	Dot plot, box plot, or scatter plot	Show units; inspect outliers and sample size
Explore geography	Country bars or map	Use maps only if geographic fields are meaningful and sufficiently complete
Understand customer ratings	Distribution or category bars	Show rating scale and sample coverage

05 | Interaction and navigation

- Keep the main filters in a predictable position and label each one clearly.
- Display the currently selected year, category, or manufacturer where practical.
- Use tooltips for additional detail without crowding the primary chart.
- Make reset/clear-filter behavior obvious if supported by the chosen platform.
- Arrange dashboard/story pages in a logical sequence from overview to deeper comparison.

06 | Design risks and controls

Risk	Design control
Too many charts	Prioritize the stakeholder questions and split content into focused views.
Misleading comparisons	State period, denominator, units, and missing-data rules.
Incomplete dataset	Hide unsupported analysis or mark it unavailable; never fabricate fields.
Visual overload	Use a consistent layout, restrained color palette, and clear chart hierarchy.

Design deliverable

A documented architecture, dashboard view plan, interaction principles, and visualization-to-question mapping. The design becomes the reference point for development and testing.